



DATA SPACES SUPPORT CENTRE

Version 1.5

Starter Kit for Data Space Designer

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Document Summary Information

Grant Agreement No	101083412	Acronym	DSSC
Full Title	Data Spaces Support Centre		
Start Date	01/10/2022	Duration	42 months
Project URL	www.dssc.eu		
Deliverable	Not Applicable		
Work Package	WP4		
Contractual due date	Not Applicable	Actual submission date	Not Applicable
Nature	Not Applicable	Dissemination Level	Not Applicable
Lead Beneficiary	NUIG		
Responsible Author	Rafiqul Haque		
Contributions from	BDVA, FIWARE Foundation, Fraunhofer-Gesellschaft, Gaia-X, IDSA, KU Leuven, NUIG, TNO, VTT		



This project has received funding from the European Union Digital Europe Programme under Grant Agreement № 101083412.

More information available at <https://dssc.eu/>

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1 Data Spaces 101

Data spaces facilitate data sharing and exchange of data among diverse stakeholders who are keen to collaborate. The [European strategy for data](#) seeks to accelerate the development of the European data ecosystem and data economy, aiming to harness the societal value of data while ensuring Europe's global competitiveness and data sovereignty. The European Union (EU) is investing in various initiatives related to [Common European Data Spaces](#), with a focus on sector-specific and domain-specific initiatives. A common European data space integrates relevant data infrastructures and governance frameworks to streamline the pooling and sharing of data resources. This **Starter Kit** is aimed at emerging data spaces that stand to benefit from a growing community of experts developing data-sharing technologies, standards, regulatory frameworks, governance models, and innovative business strategies.

A data space is *an interoperable framework based on common governance principles, standards, practices, and enabling services, designed to facilitate trusted data transactions between participants*. It is implemented by one or more infrastructures and supports one or more use cases. A data space should be generic enough to accommodate a broad spectrum of use cases. For instance, industrial data spaces can facilitate varying levels of trusted and secure sharing and trading of commercial data products, with automated and robust controls for legal compliance and remuneration. Within data spaces, compliance with regulations concerning private data is ensured, enabling data subjects and holders to maintain control over their data and its subsequent use.

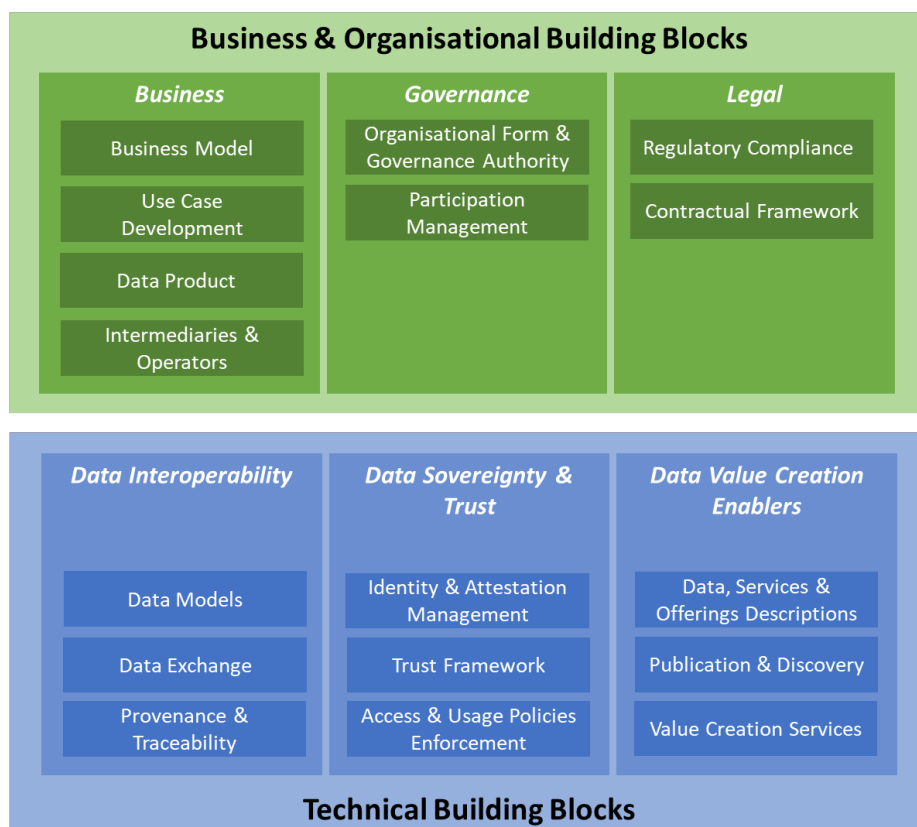


Figure 1. Building Block Overview

There are numerous fundamental technical, organisational, legal, and commercial challenges in developing, deploying, and sustaining data spaces to support data ecosystems. This Starter Kit, developed by the Data Spaces Support Centre (DSSC), offers an entry point for individuals or organisations aiming to establish or participate in a data space. This includes data product owners, data providers, data recipients, data space intermediaries, and providers of data-driven services and business applications. It offers a comprehensive overview of the **business and organisational**, and **technical building blocks**, categorised into the following pillars: business, governance, legal, data interoperability, sovereignty and trust, and value creation enablers. The updated Building Block Overview is shown below in Figure 1.

The Starter Kit helps clarify the role of the DSSC as foundational support for the European strategy for data, while pointing to a set of resources that offer further guidance and assistance. It forms an integral part of DSSC's [asset model](#) (see Figure 2). Notably, the DSSC has adopted an **asset-based approach** to enhance understanding of data spaces, both internally and externally. By employing this approach, the DSSC aims to provide clarity on the nature and importance of data space assets. In addition to the Starter Kit, the DSSC is developing assets that are crucial for the design, development, management, and governance of data space. The asset model shown in Figure 2 serves as a visual representation, illustrating the assets and the intricate interrelationships among them.

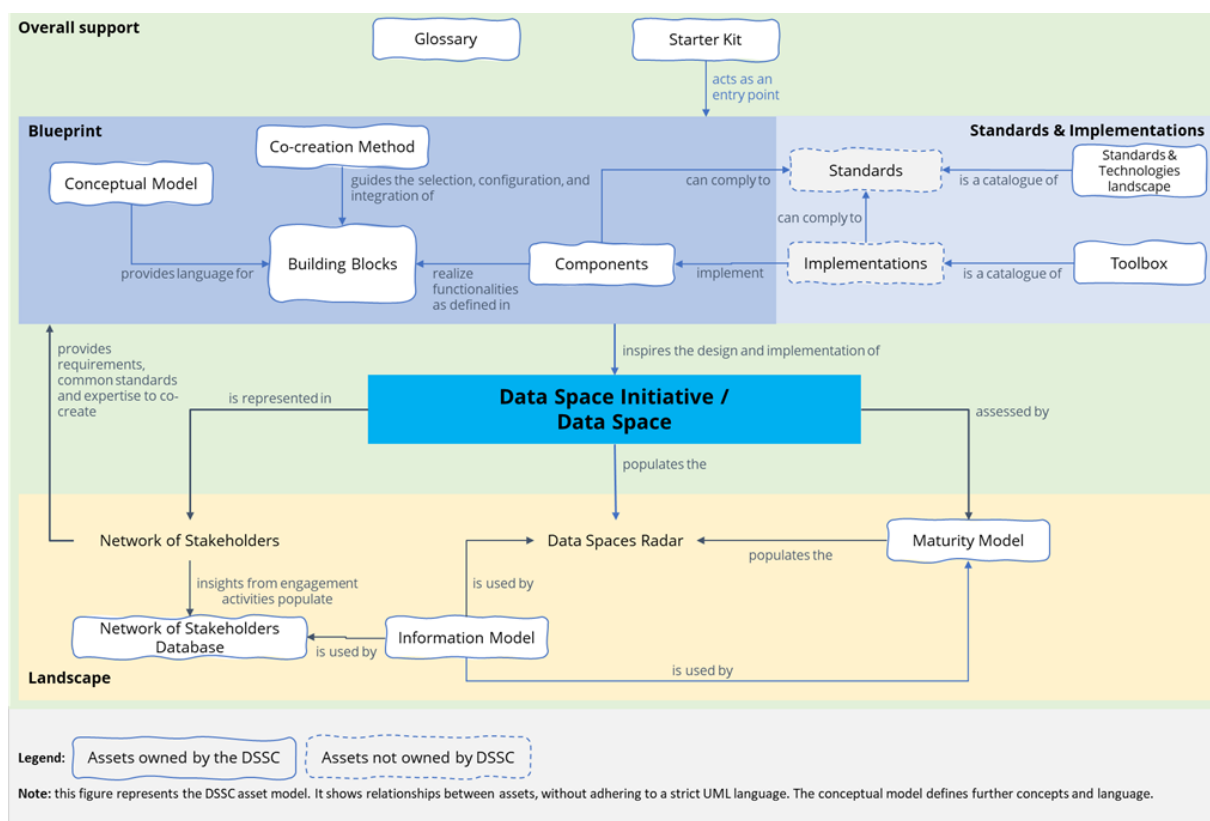


Figure 2: Data Space Support Centre Asset model

Central to the asset model is the [Data Spaces Blueprint](#), which comprises Building Blocks, a Conceptual Model, Co-Creation Method, and Components. A [Building Block](#) is a basic unit or component that can be implemented and combined with others to achieve the functionality of a data space. The [Conceptual Model](#) represents the concepts (entities) and their relationships within a data space. The [Co-Creation Method](#) is

meant to guide data spaces and data space initiatives (and their participants) through setting up, developing and operating a data space.

Spanning a broad range, from fundamental data to comprehensive standards knowledge, the asset model also includes the [DSSC Glossary](#), which provides an extensive list of data space terminology and definitions. Additionally, the asset model incorporates **Data Space Initiatives** and **Landscape**, which comprises a **Network of Stakeholders Database**, an **Information Model**, and a **Maturity Model**.

2 Data Space Start-up Checklist

Establishing a data space can pose various challenges, which many data space initiatives may find difficult to address. This Starter Kit offers a “**Quick Start**” guide by framing the journey of setting up a data space. It puts forward a set of questions, to contemplate and answer when setting up a data space. These questions are organised into five dimensions: business, legal, operational, functional, and technical. Figure 3 illustrates the essential questions that need to be addressed when establishing a data space.

Data Spaces Start-up Checklist

Business

- How does the data space create value?
- Who are the key stakeholders or participants in the data space?
- What is the business and governance model adopted by the data space?
- What are the individual and collaborative business models (Incentives) for the participants in the data space?
- What specific use cases will the data space support to create business, societal, or environmental value?

Legal

- What legal aspects must be considered when setting up a data space?
- What are the legal requirements and challenges associated with the data space?
- What legal dimensions are involved in data governance?
- How can data spaces ensure compliance with EU values and regulations?

Operational

- What is the operational governance framework for a data space?
- What daily activities and processes are crucial for maintaining the data space?
- What mechanisms are in place to facilitate data sharing and collaboration among stakeholders?
- How are data sharing agreements managed and enforced?

Functionality

- What core functionalities should a data space offer?
- What are the essential building blocks that make up each functionality?

Technical

- What formal and de facto standards should be adhered to when deploying a data space?
- How to address the heterogeneity of data?
- How to describe and publish data and services in an efficient way to ensure the findability?
- How to ensure data sovereignty and trust?
- What software requirement specifications serve as reliable references for implementing a data space?
- Which open-source software implementations are compliant with recommended standards and specifications?



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Figure 3: Data Space Check-list

The **business dimension** focuses on the value proposition, participants, use case, and business model of the data space. It addresses questions related to value creation, stakeholders and participants, business and governance models, and use cases of data space, ensuring that the data space delivers tangible value and remains financially sustainable.

The **legal dimension** covers compliance, data governance, and contractual agreements. It prompts considerations of data privacy laws, intellectual property rights, and legal responsibilities, ensuring that the data space operates within the regulatory framework and protects the rights of all stakeholders.

The **operational dimension** deals with the day-to-day management and coordination of the data space. It includes questions regarding governance structures, roles and responsibilities, and operational workflows, ensuring the smooth and efficient functioning of the data space.

The **functional dimension** examines the specific features and services offered by the data space. It addresses questions about core functionalities, ensuring that the data space meets the functional needs of its users.

The **technical dimension** focuses on the technological infrastructure and tools required to support the data space. It includes questions related to data interoperability (standardisation of the data model and common vocabulary), data sovereignty and trust, the description of data and related services, and software requirements.

It is important to note that these dimensions align with the **business and organisational**, and **technical building blocks** outlined in the latest version of the **Data Space Blueprint**. The questions concerning the business, legal, and operational dimensions relate to the business and organisational building blocks, while the functional and technical dimensions align the technical building blocks. By addressing these questions, data space initiatives will be better equipped to navigate the diverse requirements integrated into the DSSC Asset Model.

To assist data space initiatives in tackling key questions effectively, this Starter Kit offers a comprehensive overview of the five dimensions. It also proposes a structured design approach to guide the implementation of a data space. The Starter Kit delves into the specifics of each dimension, detailing their relevance and interconnections. Additionally, it explains how these dimensions align with the business, organisational, and technical building blocks outlined in the DSSC Blueprint.

3 Interoperability and Data Quality

Interoperability

Interoperability is the cornerstone of successful data spaces, enabling seamless data sharing, enhancing data quality, and fostering efficient data exchange to drive innovation and economic growth. It is essential for creating a cohesive and collaborative environment within and across data spaces where data can be seamlessly exchanged and utilised to its optimal potential. Interoperability guarantees a unified and comprehensive view of data, ensuring that information from various sources can be shared among participants in a data space without compatibility issues.

In the context of data spaces, **interoperability** operates at three distinct levels: [data interoperability](#), **intra-data space interoperability**, and **cross-data space interoperability**.

1. **Data interoperability** involves structuring and formatting diverse datasets so that they can be integrated or combined in a meaningful and efficient manner.
2. **Intra-data space interoperability** pertains to ensuring seamless data interaction within a single data space.
3. **Cross-data space interoperability** addresses the challenges of sharing data between multiple data spaces.

Data interoperability serves as the foundational level of interoperability within a data ecosystem. It encompasses core aspects such as **technical**, **syntactic**, and **semantic** interoperability. Intra- and cross-data space interoperability extends these aspects by also including **organisational**, **legal**, and **business** dimensions. Table 1 below outlines the scope and key components of these various interoperability settings.

Interoperability Aspects	Interoperability Layer		
	Data Interoperability	Intra-Data Space Interoperability	Cross-Data Space Interoperability
Technical	Technical aspects of data exchange, such as common formats, protocols, and transmission standards.	Technical aspects of data exchange within a data space including common formats, protocols, and transmission standards.	Technical aspects of data exchange between data spaces including common formats, protocols, and transmission standards.
Semantic	Common data models, vocabularies, and ontologies to provide a shared understanding of the data.	Common data models, vocabularies, and ontologies to provide a shared understanding of the data exchanged within a data space.	Common data models, vocabularies, and ontologies to provide a shared understanding of the data exchanged between data spaces.
Syntactic	Common structure and syntax of data, ensuring that data formats (like XML, JSON) are compatible.	Common structure and syntax of data, ensuring that data formats (like XML, JSON) are compatible between the systems within a data space.	Common structure and syntax of data, ensuring that data formats (like XML, JSON) are compatible between the systems and platforms across data spaces.
Organisational	Not applicable	Governance frameworks encompass policies and agreements that define roles, responsibilities, and data-sharing rules, along with the harmonisation of workflows and procedures to ensure seamless interaction between organisations participate in a data space. Additionally, they involve adherence to industry standards and regulatory requirements to ensure	Governance frameworks encompass policies and agreements that define roles, responsibilities, and data-sharing rules, along with the harmonisation of workflows and procedures to ensure seamless interaction between organisations participate in different spaces from different sectors. Additionally, they involve

		compatibility and trust within a data space.	adherence to industry standards and regulatory requirements to ensure compatibility and trust between data spaces.
Legal	Not Application	Regulations, agreements, and policy alignment collectively ensure that data sharing and usage within a data space comply with laws (e.g., GDPR, HIPAA), establish terms and conditions through legal contracts, and align organisational policies with national and international legal frameworks.	Regulations, agreements, and policy alignment collectively ensure that data sharing and usage across data spaces comply with laws (e.g., GDPR, HIPAA), establish terms and conditions through legal contracts, and align organisational policies with national and international legal frameworks.
Business	Not Applicable	Different participants within a data space establish common business strategy, business model, and business processes to achieve commonly agreed business objectives.	Different data spaces establish common business strategy, business model, and business processes to achieve commonly agreed business objectives.

Table 1: The scope and key components of data interoperability, inter-data space interoperability, and cross-data space interoperability

The **DSSC Blueprint v1.5** addresses both **data interoperability** and intra-data space interoperability within its building blocks. The data interoperability pillar consists of three technical building blocks: capabilities for defining and using shared semantics within a data space (Data Models), capabilities related to the actual exchange and sharing of data (Data Exchange), and capabilities for tracking the data-sharing process (Provenance and Traceability). Additionally, all other building blocks are developed with a focus on intra-data space interoperability.

DSSC promotes **cross-data space interoperability** by developing common terminology, models, and practices through DSSC assets (e.g., the blueprint, glossary, and co-creation method), as well as through continuous engagement with its communities (Community of Practice, Strategic Stakeholders Forum, and Liaisons and Collaborations). You can make use of the common specifications identified by the DSSC, while also making domain-specific adaptations for your particular initiative. Furthermore, we are actively working on cross-data space interoperability and identifying key collaboration models within data spaces.

Data Quality

The risks associated with low-quality data can seriously harm an organisation. Inaccurate analytics, poor customer service, and poor decision-making can all result from low-quality data, which negatively impacts the success of an organisation. The fundamental impact of data quality is the ability of the right people to access the correct data at the right time, enabling them to make informed decisions and achieve the desired results.

The key characteristics of data quality include accuracy, completeness, consistency, validity, timeliness, uniqueness, standardization, and compliance. Managing data quality involves coordinated efforts to ensure that an organisation adheres to these characteristics, which form the foundation for three types of data quality: **syntactic** (correct data formatting), **semantic** (consistent and shared data interpretation), and **pragmatic** (data being useful to its intended recipients). Data interoperability plays a critical role in data quality management. Effective data interoperability ensures that data from various sources adheres to common standards and formats, improving consistency, integrity, and availability. This standardisation reduces errors and redundancies, leading to higher accuracy and reliability of the data. Seamless data exchange facilitated by interoperability ensures that information is transferred accurately and remains intact, contributing to high data integrity. The [data model](#) and [data exchange](#) building blocks of DSSC blueprint v1.5 explain efficient approaches for standardising and exchanging data.

Accurate, precise, and complete data minimises errors and misinterpretations, enabling different systems to understand and process information effectively. Well-documented data, supported by clear metadata, provides essential context and meaning, facilitating the correct interpretation and use of data across various systems. The [data, service, and offerings descriptions](#) building block provides technical details for creating metadata using standard vocabularies. Together with the [publication and discovery](#) and [value creation services](#) building blocks of the [value creation enablers](#) pillar, it highlights the importance of quality data and discusses how they create value.

4 Business

Data spaces are a strong and collaborative alternative to platforms, creating “multi-sided business models” that rely on network effects, serving both “supply” and “demand” of data and data-related services. Offering [data products](#) attracts users of data, and a large user base attracts additional [data resources and services](#). The challenge in making this work, however, is organisational rather than technical. Creating substantial value requires consensus through collaborative efforts from multiple users and organisations.

An end-to-end development of a data space goes through the [development lifecycle](#) which consists of exploratory stage, preparatory stage, implementation stage, operational stage (including maintenance and improvement), and scaling stage. The value of data spaces, as with any multi-sided business model, is most prominent in the scaling stage. This is because the data space and its technologies and services become available to more users and organisations compared to the previous stages. The [business model](#) building blocks offer guidance and resources to assist in formulating a business plan tailored to the data space's needs. It outlines the interconnection between data products and services, illustrating how stakeholders and customers can leverage them. This integration is essential for establishing a financially sustainable business model. Data products and **value creation services** drive the creation of [use cases](#), which form the cornerstone of the business model.

In public sector-oriented data spaces, the [business building blocks](#) facilitate the assessment of financial sustainability and stakeholder value. For profit-driven initiatives, they provide insights into packaging products and services and articulating their value to customers. Across all contexts, they help to identify the key stakeholders required to ensure the viability and value of data spaces.

Beyond the key objective of secure and sovereign data sharing for value creation, data spaces can be established for different purposes. In fact, patterns of collaborative business cases can be identified. For example, a data space can reduce the overall cost of linking systems or collectively ensuring compliance with data regulations, typically in established industries (e.g., [Smart Connected Supplier Network](#) in manufacturing). It can produce joint value by creating a set of data and logic that cannot be achieved individually, as in end-to-end multi-modal mobility services (e.g., Catena-X and the Mobility Data Space). It can operate as a marketplace where the data space can be seen as a sales channel between data providers and data recipients (e.g., [Mobility Data Marketplace](#) for open mobility data in Germany). Table 2 contains a first set of business-case patterns (which are not necessarily mutually exclusive), including an indicative assignment of [Gaia-X](#) lighthouse projects.

Business-case pattern	Description	Examples
Cost Sharing	- Data space participants share their data to meet shared requirements (e.g., compliance, process efficiency, transparency). - Every member saves money and time by sharing the burden.	Catena-X SCSN
Joint Innovation	- Customer innovation can only be realized by ecosystem members working together. - No single ecosystem member has all the necessary data.	Eona-X Mobility Data Space DS4SKILLS
Combined Forces	- Ecosystem members team up to prevent a limited number of dominant market players from emerging. - No single ecosystem member has the necessary resources and commitment to do this alone.	EUProGigant
Shared Marketplace	- Ecosystem members team up to provide quality-assured, easy access to data of a domain of common interest (open data, business partner data, etc.). - Transaction costs go down for all ecosystem members	Catena-X Mobility Data Space
Greater Common Good	Public and private sector share data for a greater common, societal goal (e.g., climate protection).	Mobility Data Space

Table 2: Business-case patterns of data spaces

Principles and rules need to be set according to the business-case pattern to help govern and coordinate the contributing organizations that help a data space meet its objectives and progress through the different life-cycle stages. Following the logic of multi-sided business models, the data space can only be sustained if it creates value for the involved organisations that use data-driven applications (that also need business models).

Business-model radars for data-driven applications and data space participants (See Figure 4): The [Base/X](#) business engineering approach exemplifies how to design collaborative business models.

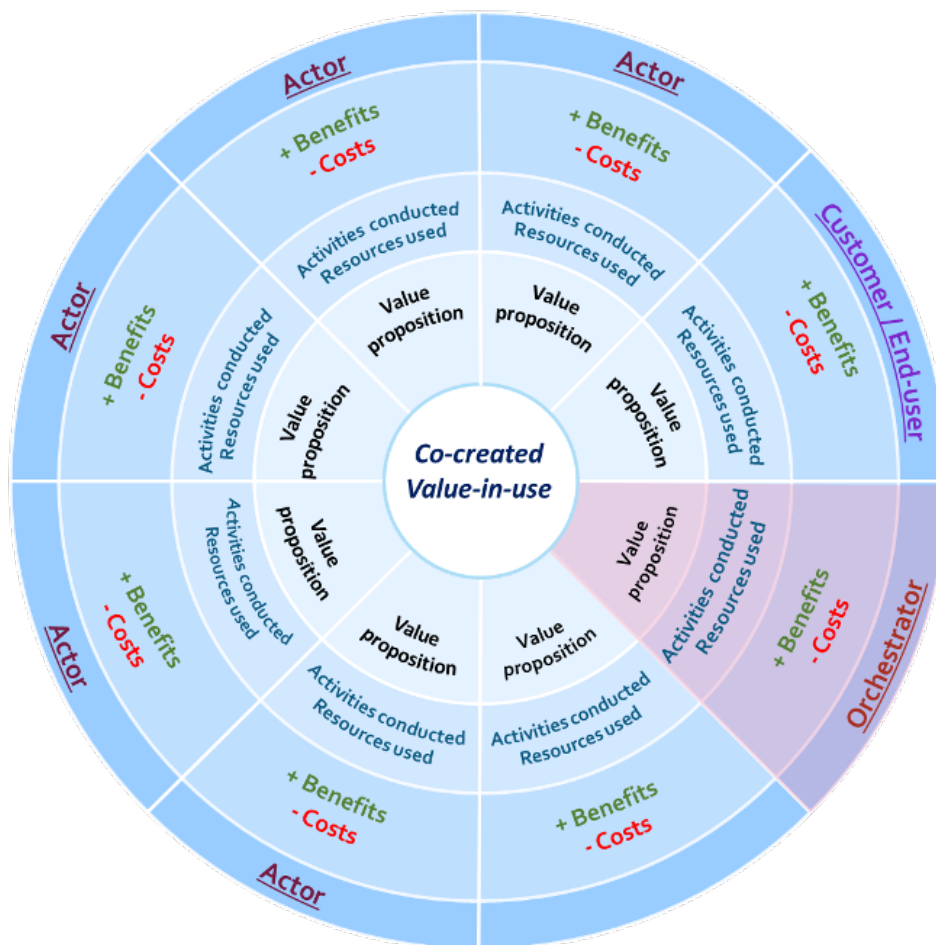


Figure 4: Business model radar

The core of this tool is the value created by the interactions of multiple organisations, including the end-user, which a single organisation cannot create alone. Catena-X, for example, represents a data space in the automotive industry and identifies the following [roles](#):

- Data recipient (called data consumer by Catena-X)
- Data provider
- Provider of business applications
- Provider of enabling services (e.g., EDC connector)
- Provider of core services (identity management, data catalogue etc.)

- Provider of on-boarding services

The Data Space for Sustainable and Smart Cities and Communities (DS4SSCC) is another example that showcases a range of different roles, as listed below:

- Cities and regions (data producer and data recipient)
- Citizens and civil society organisations (data provider and data recipient)
- Academic and research institutions (data provider and data recipient)
- Suppliers of city solutions (data provider and data recipient)

For every individual role and the entire data space, it is crucial to observe the value creation and capture, by individual participants, and the overall value created. For example, within the Smart Connected Supplier case: manufacturing companies and suppliers can exchange data with partners in their supply chain easier, faster, and more reliably. The overall value created is a governed secure data exchange for a more resilient supply chain. Such a data-driven application is one of the many similar applications in the manufacturing domain that could use an underlying data space.

Data spaces must also be managed and maintained, and consequently, a collaborative business model addressing the different organizations and roles necessary to keep the data space up and running is needed (e.g., by querying stocks and acquiring shortest time to delivery to comply rush orders of OEMs). Collaboratively, the organizations will have the above-mentioned growth and network effects and reflect the different objectives, or patterns, like cost sharing, joint value creation or marketplaces.

5 Legal

Understanding the legal aspects of data spaces can be a difficult task. To function, data spaces need to navigate a combination of well-established legislative frameworks, as well as new legislation aimed at implementing the **European strategy for data**. The evolving area of EU data and digital legislation includes requirements for data space initiatives and participants, and a variety of legal entitlements to data.

Therefore, the relationship between legal aspects and data spaces is twofold. On the one hand, data and digital legislation create a legal framework for data spaces. On the other hand, data spaces can serve as a tool for legal compliance. Data spaces can provide a valuable contribution towards the development of products and services in compliance with the law. Through the provision of policies, tools and resources, they can support data sharing in compliance with the law (for instance, with regards to personal data or intellectual property) and the development of products and services in line with legal requirements (for instance, the requirements set for the development of high-risk AI systems).

As part of the v0.5 of the blueprint, a first overview of relevant EU legislation was provided as part of the Regulatory Compliance building block (See Figure 5):

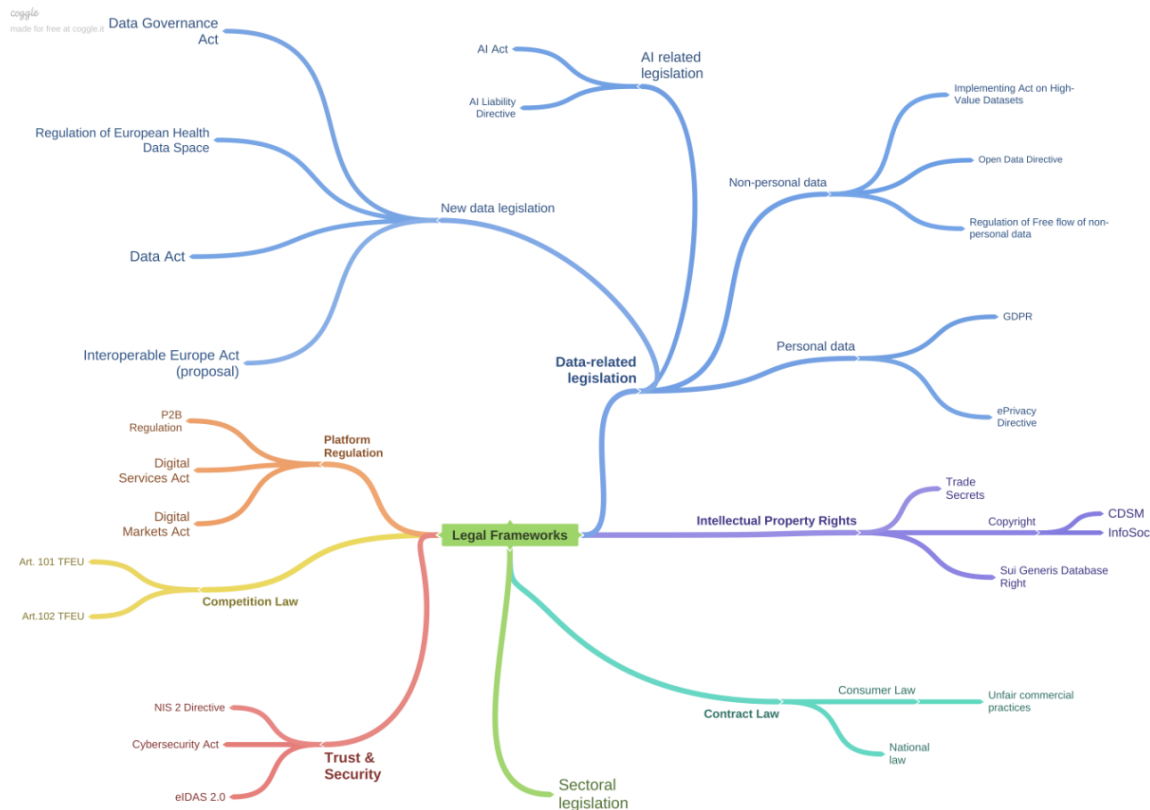


Figure 5: Relevant EU Legislations

- **Data-Related Legislation:** Data-related legislation covers various sub-categories, such as personal data, non-personal data, artificial intelligence, and sector-specific regulations governing European Data Spaces. The EU has introduced significant regulatory interventions to support the European data economy, including the [Data Governance Act \(DGA\)](#) and the [Data Act \(DA\)](#).
- **Platform Regulations:** The introduction of the [Digital Markets Act \(DMA\)](#) and [Digital Services Act \(DSA\)](#) represents targeted efforts to address specific players in the digital economy. Specifically, the DMA's objective is to regulate dominant players, particularly "gatekeepers", while the DSA is oriented towards enhancing online services' transparency, accountability, and regulatory oversight. A clearly defined ecosystem is essential for data spaces to support data sharing effectively.
- **Intellectual Property Law:** The Intellectual Property Law regime, particularly copyright and the *sui generis* database right (SGDR), is relevant to the data economy. Existing legal frameworks continue to provide legal entitlements to data through intellectual property rights (IPR). Data spaces must consider intellectual property rights to address rights allocation, protection, licensing, and commercialisation issues. Understanding and respecting IPR within data spaces fosters an environment conducive to opening up privately held data. In addition, trade secret protection is a mechanism used to safeguard confidential business information and maintain a competitive advantage.
- **Competition Law:** EU competition governs collaboration and cooperation among competitive data space participants. The law aims to protect the competition process and maximise consumer welfare by fostering market entry and innovation. Competition law ensures fair, transparent, and non-discriminatory data exchange. Depending on the circumstances, requirements set by competition

law may result in obligations to cooperate and share data or prohibitions on sharing specific data with certain entities.

- **Trust & Security:** Trust and security are fundamental to data spaces. Robust cyber resilience and organisational security measures are essential for data spaces. Such measures strengthen privacy, build trust among stakeholders and promote best practices in data security. Legal developments in digital identity, particularly online identification, authorisation and authentication, must be considered as these areas play an essential role in enabling data spaces. The eIDAS Regulation, which lays down conditions for electronic identification and trust services, is currently under revision. Trusted identification and authentication are crucial for the management of data spaces.
- **Contract Law:** Contracts aim to protect the interests of all data space participants and, therefore, assume a crucial role in the data space governance framework. Contracts foster a secure and collaborative environment for data sharing and innovation by establishing rights and responsibilities among data space participants and promoting accountability and trust. There is no harmonised contract law at the EU level, so national contract law must be considered. Contract law is presented here as a cluster, as this area of law interacts with other legal frameworks, underscoring its place in the broader legal landscape.

For a more extensive overview of EU regulation for the digital world, see [A dataset on EU legislation for the digital world](#) (published 16 November 2023).

The DSSC team continues to work with our Community of Practice to further prioritise the challenges and provide guidance on establishing a legally sound data space. The DSSC thereby builds on previous work, such as the [SITRA Rulebook for a fair data economy](#), [OpenDEI design principles](#), the [BLOFT Framework](#), the [Support Centre for Data Sharing \(SCDS\)](#) and the [European Hubs for Data \(EUH4D\)](#).

As part of this work, the DSSC developed [legal building blocks](#) as part of its Blueprint, and aims to provide a number of tools to help implement these building blocks, such as a catalogue of contractual modules.

The **Regulatory Compliance** building block provides concise information on a number of highly relevant legislative frameworks relevant to data spaces. It provides a non-exhaustive list of triggers (understood as essential elements or criteria whose existence or specific function in a given context result in the application of the legal framework). Such triggers include, for example, the processing of personal data and the provision of services of data intermediation. The legislative frameworks linked to each of the triggers are summarily described and further information is provided where it is available.

The **Contractual Framework** building block provides an overview of the most common agreements that support the establishment and functioning of a data space. It provides both an overview of different agreements related to the establishment of the data space (e.g., a constitutive agreement and general terms and conditions), as well as agreements related to data transactions. For each agreement, concise information is provided on the matters that are usually regulated through such agreements.

6 Operational Activities

The operational aspects of data spaces encompass the day-to-day activities required for their successful functioning. These activities are typically addressed within the scope of data and organisational governance design, ensuring that the data space operates efficiently and effectively.

Mutual trust is essential for the exchange of data in data spaces. Building trust relies on effective governance, which establishes the protocols for data sharing. Operational activities include establishing governance agreements that ensure compliance with relevant regulations, such as the [General Data Protection Regulation \(GDPR\)](#), [Data Act](#), and [Governance Act](#), among others. These agreements define the roles and responsibilities of participating organizations, set out the procedures for onboarding new organizations, and outline the decision-making processes.

Adherence to these governance agreements holds paramount importance for participants within data spaces. Consequently, organizational regulations or contracts should be complemented by technical solutions, and vice versa, to govern the utilization of data products by participants. Effective dispute resolution mechanisms are also put in place to handle conflicts that may arise between participants.

In essence, achieving comprehensive usage control necessitates the incorporation of control points within data spaces. The complexity of implementing such control points can vary. The [governance building blocks](#) assist in the development of activities pertaining to governance. They include policies, standards, and frameworks that guide the operation of the data space. For example, a data space may implement data access policies that define who can access what data and under what conditions. Standards for data quality and interoperability ensure that data shared within the data space is reliable and can be seamlessly integrated.

Effective governance in data ecosystems is underpinned by two essential building blocks: [Organisational Form and Governance Authority](#), and [Participation Management](#). The Organisational Form and Governance Authority building block bear the responsibility of crafting regulations, delineating the scope of governance, ensuring adherence, and resolving conflicts to nurture trust and uphold data integrity. It accentuates the necessity for adaptability and evolution, especially within intricate data landscapes or federations. Concurrently, the Participation Management building block simplifies governance procedures while identifying pivotal elements for participant engagement, thereby highlighting the perils associated with poorly managed participation. This building block emphasises the imperative need for lucid regulations, inclusive governance practices, and unwavering adherence to legal and regulatory standards. Critical components encompass participant onboarding protocols, role delineation, standard regulations, access protocols, and strategic decision-making

Beyond governance, operational activities also encompass various **business operations** that are crucial for the data space's sustainability and growth. Process streamlining and automation play a key role in enhancing efficiency. For instance, automated data processing workflows can reduce the time and effort required to manage data. Marketing and awareness activities are vital for attracting new participants and fostering a community around the data space. This can involve outreach programs, promotional campaigns, and educational initiatives to inform potential users about the benefits and functionalities of the data space.

Monitoring and logging of day-to-day data exchange between data providers and consumers is another essential aspect of operational activities. This ensures that the flow of data is monitored, tracked, and logged, which helps in detecting issues and resolving them quickly. Furthermore, the operational aspects include maintaining the technical infrastructure that supports the data space. This involves ensuring the reliability, scalability, and security of the data-sharing platform. Regular updates and maintenance of the software and hardware components are necessary to keep the system running smoothly.. In cases of core components going down unexpectedly, alerting mechanisms should be in place to ensure that issues are addressed promptly. Implementing robust cybersecurity measures is essential to protect sensitive data from breaches and unauthorised access.

Documentation in the form of user, developer, and administrator guides is also essential to support the day-to-day operations of the data space. This Starter Kit helps users to understand how to use the system and provides guidance for developers and administrators on how to manage and maintain the system. Additionally, the DSSC has established a support infrastructure including a [helpline](#) which enables data space initiatives and users to contact DSSC experts and obtain the necessary information. Continuous feedback mechanisms can help identify areas for improvement and enhance user satisfaction.

In essence, effective operational activities serve as the backbone of data spaces, facilitating seamless data exchange, promoting collaboration, and empowering organisations to derive maximum value from their **data products**. By prioritising operational excellence and continually refining processes and practices, data space initiatives can create a resilient and adaptable platform that meets the evolving needs of diverse stakeholders in today's data-driven world.

For information about operational processes including **Onboarding and offboarding, Publishing use cases, Publishing data products, Publishing data requests, Matching, Data transactions, and Repeat**, refer to the data space **operational process documentation**.

7 Functionality

In both existing and future data spaces, several building blocks (See Figure 1 in [Data Space 101](#)) must be considered during setup. These building blocks define critical areas where informed decisions are necessary to facilitate effective and trustworthy data sharing among participants. Each building block represents a foundational element that impacts the overall functionality, security, and governance of the data space. This includes aspects such as data interoperability standards, compliance with regulatory requirements, mechanisms for data access control, and protocols for data integrity and privacy. By carefully addressing these building blocks, organisations can ensure that their data space is robust, scalable, and capable of fostering collaboration among diverse participants.

An in-depth overview of the business, legal, operational activities (driven by the *business and governance building blocks*) has been presented earlier. Vital to data spaces is a robust technical foundation, characterised by the technical building blocks (See Figure 1 in [Data Space 101](#)). These building blocks are organized into three primary pillars, as outlined below:

Data Interoperability: Data spaces should provide a solid framework for efficient [data exchange](#) among participants, supporting the complete decoupling of data providers and consumers, making data services FAIR

([Findable, Accessible, Interoperable and Reusable](#)). Interoperability between domain-specific data spaces is crucial for two reasons. First, an individual or organisation is never just part of one single space but operates in different data spaces simultaneously. If different technologies are adopted for interoperability, data spaces will turn into silos. Therefore, interoperability helps provide users with more value by reducing integration and adaptation costs. It is also crucial to achieve the vision of the European strategy for data, to create one common data space, comprising many sector-specific data spaces. Second, use cases are not limited to a single data silo. Fragmentation of the data economy must be prevented to reap the maximum value for organisations and individuals. This requires the adoption of a “common lingua” every participant uses, which materialised in adopting interoperable interfaces for data exchange and the definition of compatible [data models](#). Interoperable mechanisms for the traceability of data exchange transactions and [data provenance](#) are also needed.

[Data Sovereignty & Trust](#): The Data Sovereignty and Trust building blocks constitute essential technical mechanisms aimed at ensuring the reliability and authenticity of participants' information within a data space, thereby fostering trust among them during data interactions. These building blocks serve to guarantee the authenticity and integrity of participants' information, fostering trust during data transactions. Anchored by a framework grounded in common standards and agreed-upon policies, these building blocks mitigate lock-in effects for users, adhere to FAIR principles, and incorporate verification and authentication mechanisms to bolster security. The three key building blocks—[Identity and Attestation Management](#), [Trust Framework](#), and [Access and Usage Policies](#) Enforcement—play integral roles in data space governance. Identity and Attestation Management ensures the reliability of participant information, while the Trust Framework verifies adherence to established rules and standards, promoting trust among participants. Access and Usage Policies Enforcement allows for the effective governance of participant behaviour within the data space, ensuring integrity and security throughout data interactions. Together, these building blocks establish a robust foundation for data sovereignty and trust within data spaces, ensuring the integrity and reliability of the ecosystem as a whole.

[Data value creation enablers](#): The primary objective of a data space is to foster value creation among participants by facilitating data sharing. This entails participants being informed about available data, services, and offerings within the data space, understanding how to access and utilize them, and engaging in relationships that involve potential data transactions while ensuring proper usage. The technical foundation supporting these objectives is comprised of three key building blocks. Firstly, the [Data, Services, and Offering Descriptions](#) block equips data providers with tools to accurately and comprehensively describe data, services, and offerings in a manner understandable by all participants, including related data policies. Secondly, the [Publication and Discovery](#) building block enables data providers to publish descriptions of their data, services, and offerings, ensuring they can be discovered by potential users following the FAIR principles (Findable, Accessible, Interoperable, and Reusable) as much as possible. Finally, the [Value Creation Services](#) building block complements the data space's enabling services by providing additional capabilities aimed at enhancing the value derived from data transactions, facilitating various essential tasks, processes, or functions required for users' operations, directly or indirectly.

For each building block, a data space can make specific choices. For example: how certain bodies provide ‘trust’ in the ecosystem, based on what criteria, what are the necessary steps and the extent to which marketplace functionalities will be arranged will vary from one domain to the other. However, the underlying fundamental specifications and standards to define those can be the same. There can also be best practices

and technology options. This is where the DSSC seeks to achieve synergies: allowing participants of one data space to connect to others: by using similar, compatible specifications, standards, and technologies for each of the building blocks. Ultimately this should allow for seamless ‘roaming’ of participants from one data space to the other.

8 Technology

The functionality of data spaces is implementation-agnostic. However, for data space initiatives, it is crucial to understand not only the necessary functionalities but also the available standards and software implementations that can support them. Adopting common standards and software components is essential for enhancing efficiency, ensuring transparency, and driving growth. This approach promotes interoperability, simplifies integration, and fosters a more collaborative and innovative data ecosystem, ultimately leading to more effective and scalable data spaces.

Furthermore, leveraging common standards not only reduces the time and cost associated with custom integrations but also mitigates risks related to data inconsistencies and incompatibilities. Also, the use of established software components can accelerate development cycles, allowing designers to focus on adding value through unique functionalities rather than reinventing the wheel.

The DSSC will not develop components itself but will provide a comprehensive landscape of contributions to recommend compatible technology stacks. These contributions include:

- Identifying and recommending essential standards that should be adhered to for consistency and interoperability. This includes internationally recognised standards as well as widely accepted de-facto standards that have emerged from industry practice.
- Addressing gaps where standards are lacking, resolving optional parts or deficiencies in existing standards, and detailing how different standards can be integrated. These specifications will serve as crucial references for implementations. They will provide detailed guidelines on how to fill in the gaps, ensuring that all aspects of data sharing and integration are covered, and that various standards can work together seamlessly.
- Recommending a variety of software implementations, including products, projects, and repositories, developed by different implementers to foster a diverse and robust ecosystem. By highlighting open-source solutions, the DSSC aims to encourage transparency, collaboration, and innovation within the community. These implementations will serve as practical tools for developers and organisations looking to adopt or contribute to the data space infrastructure.

The DSSC has compiled and analysed an extensive inventory of standards, specifications, and implementations. This compilation is systematically structured according diverse categories and building blocks. Currently, the following categories of standards and technologies have been carefully curated and made available:

- [Data Interoperability standards and technologies landscape](#)
- [Data Sovereignty and Trust standards and technologies landscape](#)
- [Data Value Creation standards and technologies landscape](#)

The building blocks relevant to the standards and technologies within these categories are selected based on rigorous criteria such as compatibility, adoption, and usability (refer to the [Collection of Standards and](#)

[Technologies landscape](#) for further details). The purpose is to demonstrate how different standards and technologies can seamlessly complement each other while actively avoiding vendor lock-ins and proprietary software. This approach aims to cultivate a more open, adaptable, and interoperable data space ecosystem, empowering participants to freely select and integrate the most suitable technologies

The DSSC is currently developing the **DSSC Toolbox**, a curated catalogue of implementations including reference implementations. This toolbox will serve as a comprehensive repository to organise, store, and discover various implementations of data space components. It will provide detailed metadata for each implementation, including information about the maintainers, the specific data space component being implemented, and additional relevant details. This resource aims to facilitate easy access and utilisation of data space components, promoting standardisation and interoperability within the data space ecosystem.

The DSSC will release different versions of building blocks (the current version being [DSSC Blueprint Version 1.5](#)) and will propose “releases” of compatible technology stacks to account for evolving business requirements and technological advancements. The proposed implementation of new releases for existing technical building blocks, as well as new potential technical building blocks, will be validated according to a validation schema to assess the DSSC criteria for data space components.

9 A Design Approach for Data and Organizational Governance for Data Spaces

This section provides an overview of the operational mechanisms of the design approach, rooted in the data and organisational governance of a data space. The design approach outlines the essential building blocks, standards, and services needed to implement data space functionalities, offering key insights into design decisions for both technical and business and organisational building blocks. The DSSC is developing [design principles](#) to help data space initiatives make informed design decisions.

Additionally, the DSSC has introduced the [Co-Creation Method](#), which includes [Development Processes](#) and [Operational Processes](#), to guide data spaces and their participants through the stages of setting up, developing, and operating a data space. The Co-Creation Method offers a practical, hands-on approach to integrating building blocks during the construction of a data space. It supports data space initiatives throughout the [development cycle](#) and is designed to accommodate initiatives at different levels of maturity.

The development processes occur during the first three stages of the development cycle: **exploratory**, **preparatory**, and **implementation** (refer to Figure 6). These processes can also be applied in a modular way, meaning that a data space or data space initiative can select specific development processes—or portions of them—based on the circumstances. The development processes are linked to the building blocks specified in the blueprint, which provide the fundamental elements needed to assemble the data space. The fundamental pieces or elements needed to assemble the data space are provided by the building blocks. The 1.0 version of the blueprint's development processes shed light on the proper sequencing for these building blocks. Finally, the set-up of the data space designed within the development processes should allow the operational processes within the data space to function properly.

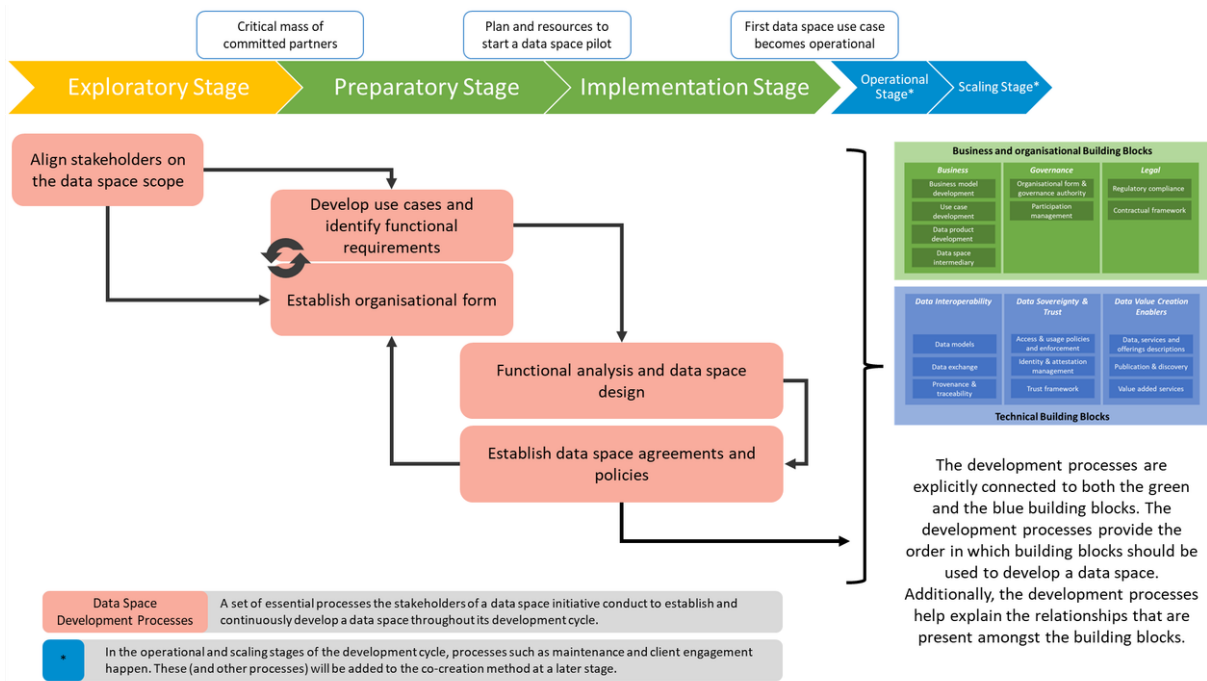


Figure 6. Development processes and the relation to the building blocks

A data space should be able to carry out multiple tasks once it is operational. Maintenance of particular components of the data space, participant onboarding, and use case publication are examples of operational procedures. During the exploratory, preparatory, and implementation phases of the development cycle, the operational processes are designed. Through the development processes, the technical elements needed to run the operational processes are created and put into place. To enable the operation of the processes, governance structures and legal documents are designed and implemented in addition to the technical parts and business models. An illustration of an operational process is shown in Figure 7.

Eight separate operational processes are shown here, which when combined make up the crucial participant flow. The critical participant flow is an essential collection of processes, as it explains the most important operational processes the data space should facilitate for its participants. Detailed information on the critical client flow and its operational processes can be found [here](#).

In the context of data spaces, data governance is intrinsically linked to the governance of the involved organisations and individuals. The [literature](#) provides a synthesis of existing works and offers insights into the various factors and decisions involved in setting up governance. However, the establishment of these governance structures must be undertaken by the organisations responsible for creating and maintaining the data space. Two approaches have proven particularly effective in designing data space governance: [The data-sharing “use case blueprint” \(Data Sharing Coalition\)](#), and (2) The templates for data space governance agreements from the Rulebook for a fair data economy ([Sitra](#)) and the [IDSA rulebook](#). The governance design will have to address, among many other aspects, the above-mentioned objectives of the data space and the evolution of the data space through the different life cycle stages. As new works on governance appear, regular alignment with the legal governance to come to one multidisciplinary governance structure is important.

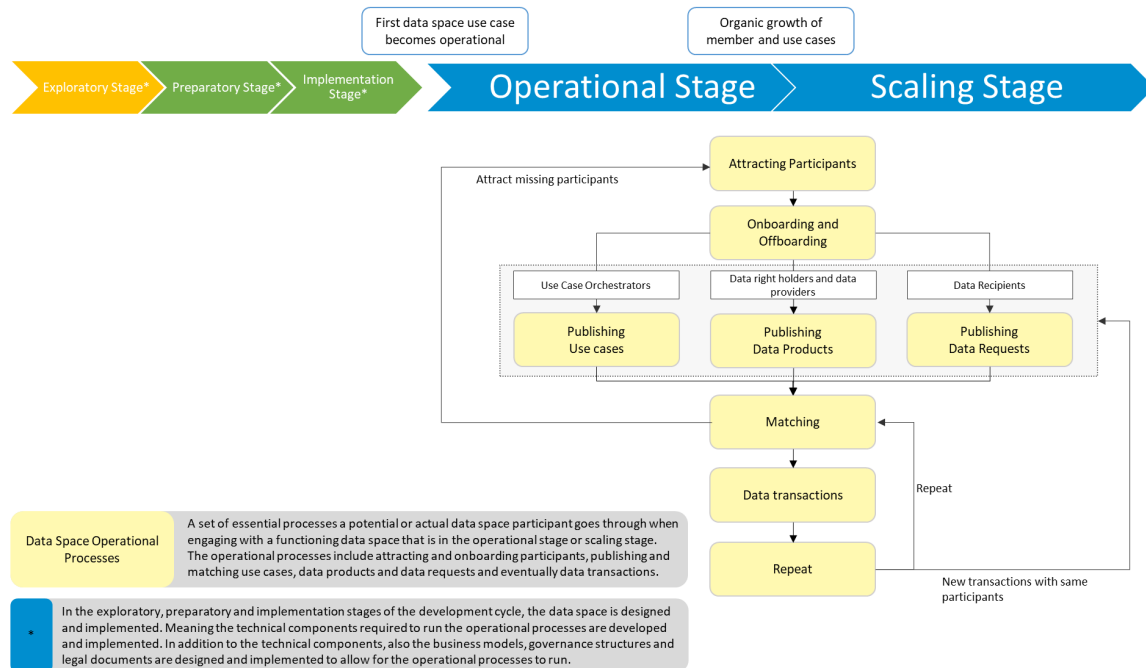


Figure 7: The critical client flow containing operational processes, mapped to the development cycle phases

In a data space, multiple organisations, each assuming different roles, must collaborate to design, implement, and maintain the data space. This collaboration necessitates a shared understanding of the purpose, objectives, and methods of working together. Governance and business models are closely interlinked, as effective governance must ensure that the data space operates cohesively, almost as if it were a single entity. Furthermore, governance is a broad topic that can encompass various aspects of a Data Space, such as legal, operational, and technical considerations, as highlighted in the [literature](#).

Governance models in the context of data-driven applications and data sharing involve multiple choices that must be made in coherence by multiple organisations. Such as data sources, access conditions, algorithms, interfaces, onboarding rules, and service levels. It is too complex for a single person to make the correct choices for all available options and expect this to yield the desired outcomes and impacts, supported by all organisations involved. To deal with this, governance modelling is approached as a design process to develop a preparatory framework of business and governance models.

Appendix 1 – DSSC Supports

The DSSC will support the data space community following a heartbeat and release cycle, creating many assets in collaboration with the Community of Practice (CoP). The Support Centre drives adoption through support activities, a platform <http://dssc.eu> for knowledge and asset sharing, a help desk, toolboxes, and active engagement with the Community of Practice (CoP). The support options are packaged to serve the needs of the data space initiatives at different maturity levels. The work of the DSSC will continuously evolve with a user-centric approach as the result of co-creation with the stakeholders and the sharing of lessons and experiences between data spaces.

Support Platform: The Support Platform is accessible through the Portal at <https://dssc.eu>. It will be enriched following the project's timeline and its assets' release. The Support Platform serves the following objectives:

The Go-to-place to get information and learn about data spaces.

Access point to join the co-creation process of the data spaces blueprint and other assets supporting the realisation of data spaces.

An entry point to organise the support by data spaces experts (in priority for data spaces funded under the Digital Europe Programme).

Get monthly highlights about DSSC activities : [Subscribe to the Newsletter](#)

Collaborate with the DSSC: [Get Involved](#)

Submit your request for contributions and provide feedback about the DSSC assets and submit an implementation for being published on the Toolbox : [Provide Feedback and Contributions](#)

For support requests please reach out via e-mail to contact@dssc.eu. The support team will be extending and tailoring their scope according to the evolution of the DSSC assets and service.

Appendix 2 – Resource Inventories

The resource inventories provide an overview of both the *state-of-the-art* and the *state-of-the-practice* in data spaces. Within this initial set of resources, we have taken an inclusive approach to showcase the breadth of ongoing work. The inclusion of a resource does not infer an official endorsement from the DSSC. Resources will be added and removed over time.

DSSC Resources

Name	Description	Source
Conceptual Model	The conceptual model of data spaces provides a coherent set of concepts, relationships between them, and constraints that document a particular way of thinking about data spaces.	Conceptual Model of Data Spaces
DSSC Glossary	The glossary establishes a consistent and coherent terminology for DSSC communication and publications. The glossary is used on a “comply or explain” basis in DSSC work.	DSSC Glossary
Business and Organisational Building Blocks	It covers the business, governance and legal building blocks of data spaces – collectively called organisational and business building blocks. It aims at data space initiatives and participants seeking to make their systems not only financially sustainable and legally compliant but also focused on creating value and making the initiative meet its objectives.	Business and Organisational Building Blocks
Technical Building Blocks	The objective of the technical building blocks is to identify these capabilities and to provide common specifications for each of them. This allows data space governance authorities and participants to make technical choices in an informed way.	Technical Building Blocks
Co-Creation Method	The co-creation method is meant to guide data spaces and data space initiatives (and their participants) through setting up, developing and operating a data space.	Co-creation Method
Design Principle	The design principles provide essential guidelines to support data space initiatives in making well-informed design decisions tailored to the target data space.	Design Principles

External Resources

Name	Description	Source
Data Spaces 101		
Designing Data Spaces	It provides a comprehensive view of data ecosystems and platform economics, from methodical and technological	Fraunhofer

	foundations to reports from practical implementations and applications in various industries. ¹	
Data Spaces: Design, Deployment, and Future Directions	Data sharing and exchange techniques using data spaces. Theory, technologies, methodologies, and best practices. ²	BDVA
Data sharing Canvas	The Data Sharing Canvas is the foundation for generic and harmonised agreements which, once implemented, enable data sharing at scale within and across domains and sectors. This has been created together with the 40+ Data Sharing Coalition participants from different domains that represent over 100.000 organisations. ³	Data Sharing Coalition (DSC)
FIWARE for Data Spaces	This white paper brings a perspective on how data spaces enabling the trusted and effective exchange of digital twin data between smart applications can be achieved. Furthermore, it brings the first perspective on how FIWARE and IDS Reference Models can be reconciled and contribute to accelerate the materialisation of Gaia-X. ⁴	FIWARE
Gaia-X Framework	Gaia-X aims to connect the Data and Infrastructure Ecosystems and relies on 3 conceptual pillars to achieve that: -Gaia-X Compliance: Decentralized services to enable objective and measurable trust -Data Spaces / Federations: Interoperable & portable (Cross-) Sector data-sets and services -Data Exchange: Anchored contract rules for access and data usage In concrete terms, for each of these pillars there are 3 types of deliverables: Functional specifications, Technical Specifications and Software (Gaia-X Framework).	Gaia-X
Gaia-X Clearing Houses	Gaia-X Clearing Houses (GXDCHs) are nodes of verification of the Gaia-X rules. It is the place to go to obtain Gaia-X	Gaia-X

¹<https://link.springer.com/book/10.1007/978-3-030-93975-5>

²https://link.springer.com/chapter/10.1007/978-3-030-98636-0_1

³<https://datasharingcoalition.eu/app/uploads/2021/04/data-sharing-canvas-30-04-2021.pdf>

⁴<https://www.fiware.org/marketing-material/fiware-for-data-spaces/>

	compliance and become part of the Gaia-X ecosystem. (GXDCH - Gaia-X: A Federated Secure Data Infrastructure)	
How to build Data Space	The knowledge base (a.k.a. "How to build Data Spaces?") is the store of information that relies on IDS expertise that is meant to support building IDS components and contribute to the existing open source components based on a five-step approach. ⁵	IDSA
Principals for Data Spaces	It defines the design principles for data spaces, agreements on the building blocks for a soft infrastructure and governance for data spaces. ⁶	OPEN DEI
Real-time Linked Dataspaces	The Real-time Linked Dataspace (RLD) is an enabling platform for data management for intelligent systems within smart environments that combines the pay-as-you-go paradigm of dataspace, linked data, and knowledge graphs with entity-centric real-time query capabilities. ⁷	Insight
European-Governed data sharing space	It enables data exchange and unlocking AI potential to create the conditions for developing a trusted European data-sharing framework. ⁸	BDVA
The European Common Data Space	Data.europa.eu and the European Common Data Spaces (ECDS). ⁹	EC
Data spaces landscape	Data spaces landscape is an overview and relations of data spaces in initiatives, standards, and tools. (public available end of February)	IDSA
Gaia-X Trust framework	The Gaia-X Trust Framework is the set of rules that define the minimum baseline to be part of the Gaia-X Ecosystem. Those rules provide a common governance and the basic level of interoperability across individual ecosystems while letting the users in full control of their choices. The Gaia-X Trust	Gaia-X

⁵<https://docs.internationaldataspaces.org/knowledge-base/>

⁶<https://design-principles-for-data-spaces.org/>

⁷ <http://dataspaces.info/>

⁸https://www.bdva.eu/sites/default/files/BDVA DataSharingSpaces PositionPaper V2_2020_Final.pdf

⁹https://internationaldataspaces.org/wp-content/uploads/dlm_uploads/EN_data_europa_eu_and_the_European_common_data_spaces_0.pdf

	Framework operationalizes the requirements as defined by Gaia-X, e.g., in the Policy Rules and Labelling Document, or the Architecture Document (Gaia-X Trust Framework - 22.10 Release)	
Business: Value and Models		
Data Spaces Brochure 2021	Use cases are cross-company business processes enabled by the IDSs standard. They help identify, analyse, and evaluate user requirements for IDS. ¹⁰	IDSA
Fair Data Economy Rulebook	Designed to guide forming of trust-based data-sharing networks with a common mission, vision, and values. ¹¹	SITRA
New Business Models for Data Spaces Grounded in Data Sovereignty	This paper applies frameworks and methods, including the business model and business ecosystem canvases, to the IDS perspective. These support a structured approach and a checklist for business planning purposes. ¹²	IDSA
Legal Landscape and Governance Frameworks		
Analytical report on EU law applicable to sharing of non-personal data	A report focusing on a thorough analysis of EU legislation applicable to sharing non-personal data, aiming to provide a structured overview of all the relevant European instruments within this field. ¹³	SCDS
Digitranscope: the governance of digitally transformed society	Full report of the Digitranscope research project, exploring the increasing awareness of the strategic importance of data and emerging governance models to distribute the value generated more equitably in society. ¹⁴	EC JRC
EU regulation builds a fairer data economy	Working paper that summarises the European Data Strategy and the new legislative proposals for the data economy (Data Governance Act, Digital Markets Act, Digital Services Act,	SITRA

¹⁰https://internationaldataspaces.org/wp-content/uploads/dlm_uploads/220812_Use-Case-Bro_2022_35-MB.pdf

¹¹<https://www.sitra.fi/en/publications/rulebook-for-a-fair-data-economy/>

¹²<https://internationaldataspaces.org/wp-content/uploads/IDSA-Position-Paper-New-Business-Models-sneak-preview-version.pdf>

¹³<https://eudatasharing.eu/legal-aspects/report-eu-law-applicable-sharing-non-personal-data>

¹⁴<https://publications.jrc.ec.europa.eu/repository/handle/JRC123362>

	Artificial Intelligence Act and Data Act). It complements this overview with an exploration of the proposals' opportunities from the perspective of the public sector, SMEs, and individuals. ¹⁵	
IDSA Rule Book	This book provides a common governance framework that specifies the functional, technical, operational and legal agreements that structure roles and interactions within and across the various parts of the IDS ecosystem. ¹⁶	IDSA
Principles for a Data Economy: Data Transactions and Data Rights	Set of transnational Principles that can facilitate the drafting of model agreements or provisions to be used voluntarily by parties in the data economy. ¹⁷	ALI-ELI
Rulebook for a fair data economy	Guide for creators of fair data economy networks, providing agreement templates and other tools to facilitate building and joining data networks. ¹⁸	SITRA
White Paper on the Data Act	The data act white paper attempts to provide a first detailed analysis of the various provisions of the data act in light of the broader EU data economy policy and regulatory landscape. ¹⁹	KU Leuven
White Paper on the Data Governance Act	The white paper offers an academic perspective to the discussion on the Data Governance Act proposal ("DGA proposal"), as adopted by the European Commission in November 2020. It contains a legal analysis of the DGA proposal and includes recommendations to amend its shortcomings. The White Paper aims to cover the full spectrum of the DGA proposal and therefore offers an in-depth analysis of its main provisions. ²⁰	KU Leuven

¹⁵<https://www.sitra.fi/app/uploads/2022/06/sitra-eu-regulation-builds-a-fairer-data-economy.pdf>

¹⁶https://internationaldataspaces.org/wp-content/uploads/dlm_uploads/IDSA-White-Paper-IDSA-Rule-Book.pdf

¹⁷https://www.europeanlawinstitute.eu/fileadmin/user_upload/p_eli/Projects/Data_Economy/Principles_for_a_Data_Economy_Final_Council_Draft.pdf

¹⁸<https://www.sitra.fi/en/publications/rulebook-for-a-fair-data-economy/>

¹⁹<https://www.law.kuleuven.be/citip/en/news/item/white-paper-data-act>

²⁰https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3872703

Gaia-X Compliance	Set of rules to have legally relevant, measurable and comparable proofs of adherence to transparent and verifiable information in order to create trust at scale ²¹	Gaia-X
Functionality and Technology: Blueprints and Building Blocks		
Data Sharing Coalition	The Data Sharing Coalition builds on existing data-sharing initiatives to enable data sharing across domains. By enabling multilateral interoperability between existing and future data-sharing initiatives with data sovereignty as a core principle, parties from different sectors and domains can easily share data, unlocking significant economic and societal value	DSC
Data Connectors	This report captures 16 Data Connector variants and provides insights into their current development and usage status. Beyond the Data Connectors, this report provides insights into emerging technologies and concepts in data spaces and fundamental technologies that form a basis for data spaces. ²²	IDSA
Data Usage Control in the IDS	This paper focuses on data usage control and data provenance, which are conceptual and technological solutions to cope with data sovereignty challenges. It presents three approaches researched and developed within Fraunhofer: The MYDATA control technologies, the logic-based usage control and degree. ²³	IDSA
FIWARE for Digital Twins	It describes how smart applications from multiple domains can be developed based on the Digital Twin paradigm using FIWARE software building blocks. Central in the vision, the NGSI-LD API is proposed as an open standard API for getting access to digital twin data and the use of standard data models is promoted to ensure the portability and replicability of solutions. ²⁴	FIWARE

²¹<https://docs.gaia-x.eu/>

²²<https://internationaldataspaces.org/download/35299/>

²³https://internationaldataspaces.org/wp-content/uploads/dlm_uploads/IDSA-Position-Paper-Usage-Control-in-the-IDS-V3..pdf

²⁴<https://www.fiware.org/marketing-material/fiware-for-digital-twins/>

iSHARE	iSHARE is a coherent model ("Trust Scheme") of functional, technical, and legal agreements and standards used in the Dutch transport and logistics sector to exchange data. ²⁵	iSHARE
IDS Reference Architecture Model 4.0	The IDS RAM contains the conceptual level, including technology-agnostic specifications. The general outline of the IDS RAM is based on the five layers and the three perspectives. Each layer should reflect the main components and aspects of the IDS. ²⁶	IDSA
Minimum Viable Data Space (MVD)	A minimum viable data space (MVDS) is a combination of components to initiate a data space with just enough features to be usable for secure and sovereign data exchange, as specified by IDSA. ²⁷	IDSA
OPEN DEI building blocks catalogue	Repository on GitHub to provide a state of the art of the various building blocks from projects and data-sharing initiatives participating in the OPEN DEI survey. ²⁸	OPEN DEI
Smart Data Models	Smart Data Models is a collaborative program to provide data models for digital twins and data spaces. The SDMs are free and open-licensed, multisector, based on real use cases and adopted open standards, collaborative, at market speed, customisable to local needs and compatible with linked data. ²⁹	FIWARE
Technical Convergence Discussion Document	The discussion document is an agile paper that defines a common reference technology framework for creating data spaces. This framework is based on the technical convergence of existing architectures and models for data spaces defined by members of the Data Spaces Business Alliance. ³⁰	DSBA
Gaia-X Architecture	Set of description models and operating rules for decentralized cross-dataspace interoperability. ³¹	Gaia-X

²⁵<https://ishare.eu/>

²⁶https://github.com/International-Data-Spaces-Association/IDS-RAM_4_0

²⁷<https://docs.internationaldataspaces.org/knowledge-base/mvds>

²⁸<https://github.com/International-Data-Spaces-Association/Building-Blocks-Catalog>

²⁹<https://smartdatamodels.org/>

³⁰<https://data-spaces-business-alliance.eu/dsba-releases-technical-convergence-discussion-document/>

³¹<https://docs.gaia-x.eu/>

DCAT-AP	The DCAT Application profile for data portals in Europe (DCAT-AP) is a specification based on W3C's Data Catalogue vocabulary (DCAT) for describing public sector datasets in Europe. Its basic use case is to enable a cross-data portal search for data sets and make public sector data better searchable across borders and sectors. This can be achieved by the exchange of descriptions of data sets among data portals. ³²	EC
Interoperability		
European Interoperability Framework	The European Interoperability Framework (EIF) is a directive from the European Commission, adopted on 23 March 2017, providing specific guidelines for establishing interoperable digital public services.	EC
Use Cases		
Catena-X	Catena-X sees itself as a rapidly scalable ecosystem in which all participants in the automotive value chain participate equally. The goal: to provide an environment for the creation, operation and collaborative use of end-to-end data chains along the entire automotive value chain. ³³	Catena-X
Data Spaces Radar	The site gives an outlook on many data space initiatives and use cases, with an indication of maturity. ³⁴	IDSA
Copernicus	Copernicus is the Earth observation component of the European Union's Space programme, looking at our planet and its environment to benefit all European citizens. It offers information services that draw from satellite Earth Observation and in-situ (non-space) data.	EC
Data Space Energy Transition	Supply chain management in the energy sector - Energy communities - optimisation of supply and demand/power grid	Green Data Hub

³²<https://github.com/SEMICeu/DCAT-AP>

³³<https://catena-x.net/en/>

³⁴<https://internationaldataspaces.org/adopt/data-space-radar>

	stabilisation - data services and products for renewable energies. ³⁵	
European Health Data Space	The goal is to ensure that natural persons in the EU have increased control in practice over their electronic health data. It also aims to ensure a legal framework consisting of trusted EU and Member State governance mechanisms and a secure processing environment. ³⁶	EC
Health Data Hub (HDH), France	The Health Data Hub (HDH) is a unique gateway to health data in France. The goal is to improve the quality of care and patient support. The HDH is a platform where pseudonymised health data from different sources is duplicated and made available. ³⁷	Health Data Hub
EUCAIM	Cancer Image Europe provides a robust, trustworthy platform for researchers, clinicians, and innovators to access diverse cancer images, enabling the benchmarking, testing, and piloting of AI-driven technologies.	EC
I4Trust Data Space Experiments	i4Trust brings a framework that enables the creation of data spaces to exchange digital twin data. Latest version of this framework released January 2023 aligns with recommendations from the DSBA Technology Convergence discussion document. Thirty pioneer use cases will be experimented with in domains such as cities, mobility, energy, agrifood, logistics, ports or manufacturing. ³⁸	FIWARE, iSHARE
deployMDS	deployEMDS is a project co-funded under the EU Digital Europe Programme and responds to its outlined challenges. The project will help make the common European mobility data space a reality.	EC
Mobility Data Space	The Mobility Data Space is the data-sharing community for anyone who wants to make mobility more eco-, user-friendly, safer, and fair. ³⁹	DRM

³⁵https://gaia-x.eu/wp-content/uploads/files/2021-06/Gaia-X_Data-Space-Energy_Position-Paper.pdf

³⁶<https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52022PC0197>

³⁷<https://www.health-data-hub.fr/>

³⁸<https://i4Trust.org>

³⁹<https://mobility-dataspace.eu/>

Resilience and Sustainability Data Space	The Resilience and Sustainability Data Space will allow for transparency of the supply chain by early identification of crisis and their effects on supply and production conditions for each individual participant. ⁴⁰	Advaneo RSDS
Smart Connected Supplier Network (SCSN)	Smart Connected Supplier Network (SCSN) is a data standard that makes exchanging information in the supply chain more efficient, allowing companies to share data more easily, quickly, and reliably. ⁴¹	TNO
TEHDAS - European Health Data Space	The goal is for future European citizens, communities and companies to benefit from secure and seamless access to health data regardless of where it is stored. ⁴²	Sitra
Building Blocks	The Building Blocks are standards-based open and reusable digital solutions that enable basic capabilities, such as trusted authentication and secure data exchange. They offer basic capabilities that can be used in any European project to facilitate the delivery of digital public services across borders. ⁴³	EC
AgriDataspace	Agdatahub brings together public and private players in the agricultural sector to provide agriculture with a shared and sovereign technological infrastructure to guarantee the development of digital agriculture in France and Europe. ⁴⁴	AgDatahub
Organisations and Associations		
BDVA	With more than 230 members all over Europe, Big Data Value Association - BDVA focuses on enabling the digital transformation of the economy and society through Data and Artificial Intelligence by advancing in areas such as big data and AI technologies and services, data platforms and data spaces, Industrial AI, data-driven value creation, standardisation, and skills. ⁴⁵	BDVA

⁴⁰<https://www.resilience-sustainability-dataspace.eu/about/>

⁴¹<https://smart-connected.nl/en>

⁴²<https://www.sitra.fi/en/projects/joint-action-towards-the-european-health-data-space-tehdas/>

⁴³<https://ec.europa.eu/digital-building-blocks/wikis/display/DIGITAL/Digital+Homepage>

⁴⁴<https://agdatahub.eu/en/entreprise/>

⁴⁵<https://www.bdva.eu/>

Data Space Business Alliance	Bringing together data providers, users and intermediaries, data spaces are key to driving businesses to extract value from data competitively. With its combined cross-industry expertise, resources and know-how, the Alliance drives awareness, evangelises technology, shapes standards and enables integration across industries. ⁴⁶	DSBA
DSC	The Data Sharing Coalition - DSC aims to drive (cross-sectoral) data sharing, under control of the entitled party, by realizing data sharing use cases, enabling interoperability between data spaces, and strengthening individual initiatives. It explores and defines agreements on topics such as technical standards, data semantics, legal agreements, and trustworthy and reusable digital identities. ⁴⁷	DSC
FIWARE	Together with its members and partners, FIWARE Foundation drives the definition – and the Open Source implementation – of key open standards that enable the development of portable and interoperable smart solutions and the trusted and effective exchange of data among solutions. ⁴⁸	FIWARE
Gaia-X	With Gaia-X, representatives from business, science and politics on an international level create a proposal for the next generation of data infrastructure: an open, transparent, and secure digital ecosystem, where data and services can be made available, collated, and shared in an environment of trust. ⁴⁹	Gaia-X
IDSA	With 130 member companies, the International Data Spaces Association - IDSA aims to develop a reference architecture for international data spaces IDS, including a governance model and adoption strategy. Continue to evolve IDS-based on use cases. Establish IDS as the international standard for data exchange in the economy of the future. And support certifiable software solutions and business models. ⁵⁰	IDSA

⁴⁶<https://data-spaces-business-alliance.eu/>

⁴⁷<https://datasharingcoalition.eu/>

⁴⁸<https://www.fiware.org/>

⁴⁹<https://gaia-x.eu/>

⁵⁰<https://internationaldataspaces.org/>

MyData Global	The purpose of MyData Global is to empower individuals by improving their right to self-determination regarding their personal data. ⁵¹	MyData
Sitra	Sitra is a Finnish Fund that is influential nationally and internationally and acts as a think tank, promoter of experiments and operating models and a catalyst for co-operation. They collaborate with partners from different sectors to research, trial and implement bold new ideas that shape the future. ⁵²	Sitra
Team Data Space	Team Data Spaces brings together the leading European players in data spaces from European associations, industry, and research organisations with a common vision to deliver European data spaces. ⁵³	Team Data Space

⁵¹<https://mydata.org/>

⁵²<https://www.sitra.fi/>

⁵³<https://dataspaces4.eu/>